Appendix: How smoothing the affinity matrix affects neighborhood preservation in t-SNE

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This appendix provides dataset preprocessing details and full experimental results for all three datasets: MNIST, mouse cortex, and UCI Adult. The main paper reports results on MNIST only due to space constraints. Section A presents preprocessing details. Sections B–F present the full results corresponding to each experiment in the main paper, in the same order: per-point effective perplexity analysis, effect of smoothing on local neighborhood preservation, smoothing versus increasing perplexity, sensitivity to γ and ρ, and effect of smoothing on global structure preservation.

Dataset preprocessing details

MNIST. MNIST contains images of handwritten digits ($n = 70{,}000$, $m = 784$). Pixel values are scaled to $[0, 1]$ and the data are reduced to 50 principal components before constructing t-SNE affinities, following the preprocessing recommended by Kobak and Berens [1].

UCI Adult. UCI Adult is a tabular dataset of US census data ($n = 48{,}842$, $m = 14$). Numerical variables are standardized and categorical variables are one-hot encoded before constructing t-SNE affinities.

Mouse cortex. The mouse cortex dataset contains single-cell RNA sequencing data from adult mouse cortex, grouped into 133 clusters with a strong hierarchical organization ($n = 23{,}822$). We use the same preprocessed data as Kobak and Berens [1], which includes sequencing-depth normalization, feature selection, log-transformation, and dimensionality reduction to 50 principal components.

Additional results

Full results are available at https://github.com/shirinmhx/smooth_t-SNE. The findings reported in the main paper are consistent across all three datasets.

Figure 1 shows the per-point change in effective perplexity after smoothing across all three datasets. For the top-5 conditional mass (left column), the pattern is consistent and clear across all datasets: points with sharper affinity rows receive a larger increase in effective perplexity, with Spearman correlations of 0.669, 0.801, and 0.566 for MNIST, mouse cortex, and Adult respectively. This is consistent with the analysis conducted in Section 3: sharper distributions have more room to redistribute mass, so smoothing affects them more strongly.

For the bandwidth σ_i (right column), the relationship is less consistent. For MNIST and Adult, points in sparser regions (larger σ_i) tend to receive a larger increase in effective perplexity, with Spearman correlations of 0.490 and 0.351. For mouse cortex, however, the relationship is essentially absent (Spearman $\rho = -0.012$). This may be because the mouse cortex dataset has a more complex density structure, where bandwidth does not straightforwardly reflect local sparsity in the same way as for image or tabular data.

1 Effect of smoothing on local neighborhood preservation

Figure 2 shows $NO@k$ for $k = 1, \ldots, 90$ at perplexity $\rho = 30$ for MNIST across all three datasets. Smoothed variants ($\gamma < 1$) score better at higher values of k, with the gap growing as k increases, while sharpened variants ($\gamma > 1$) score better at small k. The same crossover pattern holds on the Adult and mouse cortex datasets. For small values of k, sharpened variants ($\gamma > 1$) outperform standard t-SNE. This is because sharpening concentrates more probability mass on the nearest neighbors, giving them stronger attractive force during optimization. For $k > 10$, this behavior reverses: smoothed variants ($\gamma < 1$) consistently outperform both standard t-SNE and sharpened variants, with the gap growing as k increases. For example, $\gamma = 0.5$ reaches approximately 0.5 at $k = 50$ compared to 0.47 for standard t-SNE and 0.42 for $\gamma = 2.0$. We observed the same crossover pattern on the mouse cortex and Adult datasets, confirming that this is not specific to MNIST. This happens because smoothing redistributes probability mass toward mid-ranked neighbors, giving them more influence during optimization rather than letting the nearest neighbors dominate.

Figure 3 shows the embeddings for all three datasets. For MNIST, the effect of smoothing is most visible: clusters are more separated and digit classes that overlap in standard t-SNE are pulled apart. For mouse cortex, the overall hierarchical structure is preserved in both embeddings, with the smooth version showing slightly more separated clusters. For Adult, both embeddings look similar, which is expected: the dataset does not have strong cluster structure, so smoothing has less visible effect on the embedding layout, even though the quantitative neighborhood preservation metrics still show improvement.

Smoothing versus increasing perplexity

Figure 4 confirms across all three datasets that smoothing is not equivalent to increasing the global perplexity. In all cases, smoothed t-SNE ($\gamma = 0.7$) outperforms matched-perplexity standard t-SNE at broader k, despite having the same mean effective perplexity. This confirms that the point-dependent effective perplexities produced by smoothing lead to different and better mid-local preservation than simply increasing the global perplexity.

Figure 5 shows $NO@30$ while varying perplexity for standard and smoothed t-SNE (with constant $\gamma = 0.7$, not tuned) across all three datasets. The same pattern holds in all cases. Smoothing strongly affects which perplexity is optimal:

for standard t-SNE the best perplexity is around 25–50 depending on the dataset, while smoothed t-SNE peaks at a notably lower perplexity of around 15–25. In all three datasets, the peak NO@30 of smoothed t-SNE exceeds that of standard t-SNE across all perplexities. However, at higher perplexities smoothed t-SNE degrades more sharply, consistent with the finding that smoothing has adverse effect when the affinity rows are already relatively smooth.

Sensitivity to γ and ρ

Figure 6 shows the sensitivity results across all three datasets. For near-local preservation (left column), the pattern is consistent: sharpening improves performance and smoothing hurts it, regardless of dataset or perplexity. This confirms the finding in the main paper holds generally.

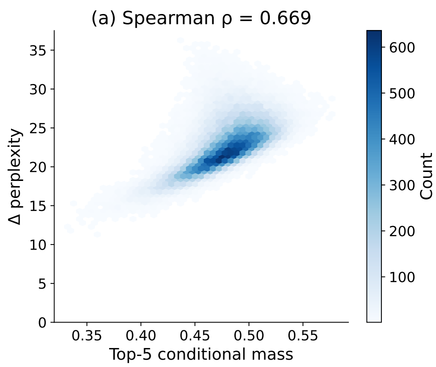
For mid-local preservation (right column), the pattern is mostly consistent for MNIST and mouse cortex, where moderate smoothing helps at lower perplexities. Adult shows a slightly different pattern: smoothing helps across all perplexity settings, with no clear shift toward sharpening at higher perplexities.

Effect of smoothing on global structure preservation

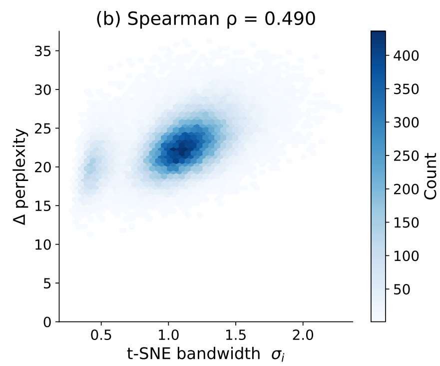
Figure 7 shows the effect on global structure preservation. The main finding holds across all three datasets: sharpening consistently degrades global structure while smoothing generally improves it, most clearly at higher perplexities. Mouse cortex shows the strongest absolute values and clearest effect, while MNIST shows the weakest effect, consistent with MNIST having strong local cluster structure that dominates the optimization regardless of γ.

References

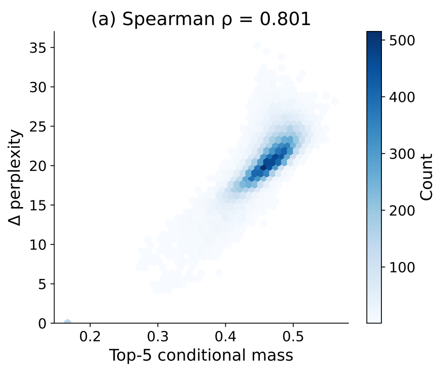
1. Kobak, D., Berens, P.: The art of using t-SNE for single-cell transcriptomics. Nature Communications **10**, 5416 (2019)



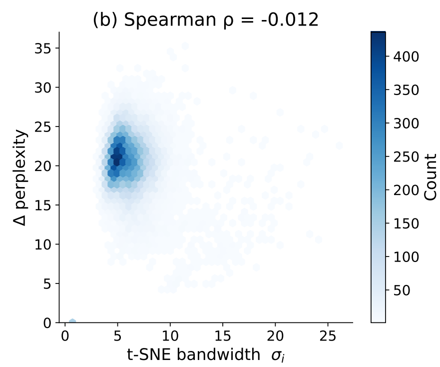
(a) MNIST



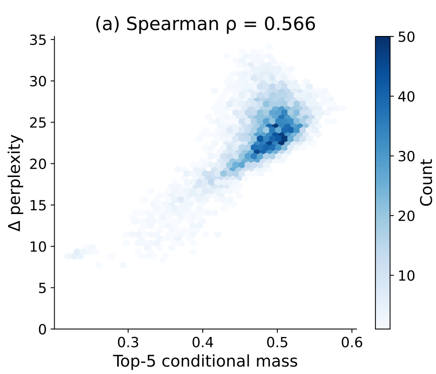
(b) MNIST



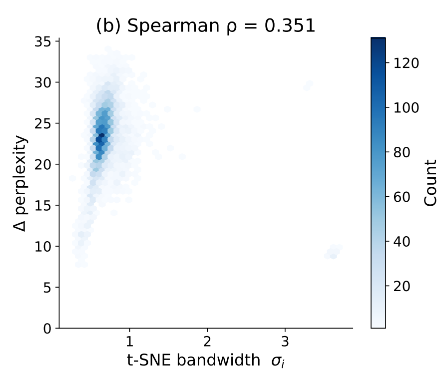
(c) Mouse cortex



(d) Mouse cortex

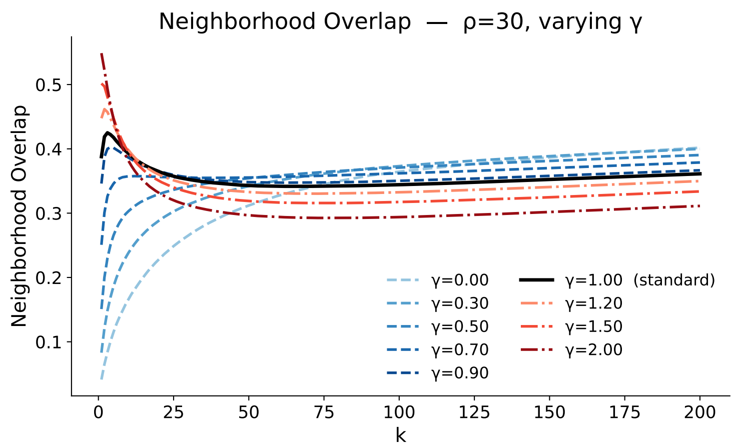


(e) Adult

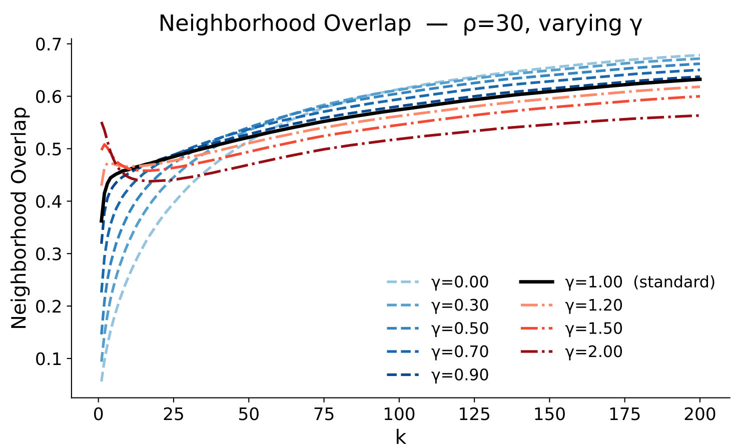


(f) Adult

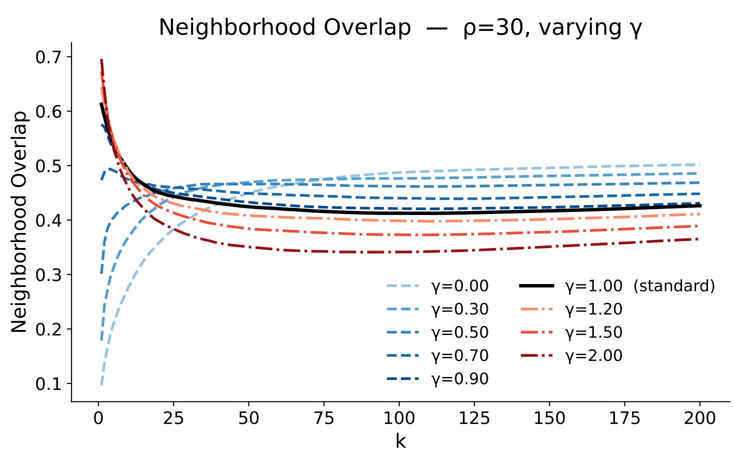
Fig. 1: Per-point change in effective perplexity after smoothing ($\gamma = 0.7$, $\rho = 30$)) (left) Versus top-5 conditional mass $M_i(5)$. (right) Versus t-SNE bandwidth σ_i across three datasets.



(a) MNIST

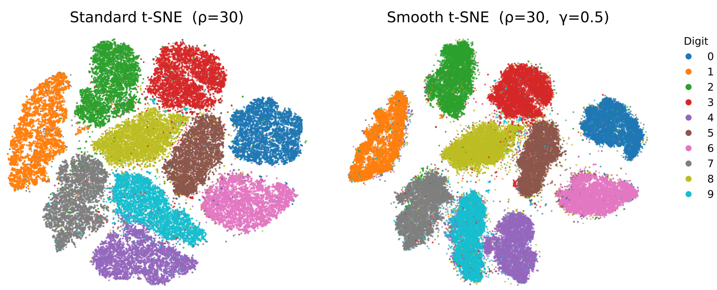


(b) mouse cortex

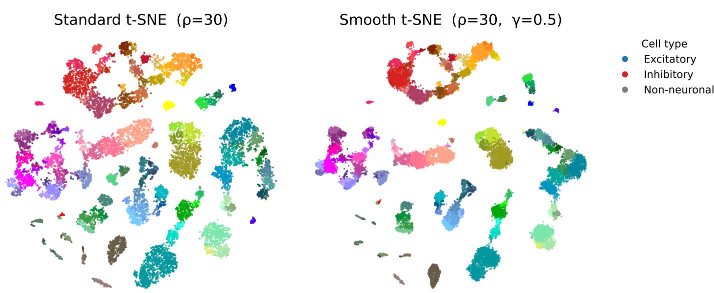


(c) adult

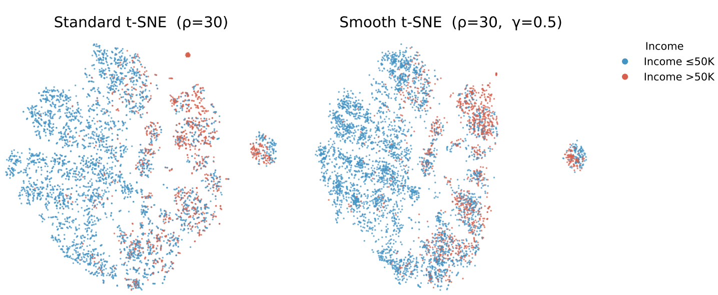
Fig. 2: $NO@k$ for $k = 1, \dots, 90$ at $\rho = 30$ with varying γ across three datasets.



(a) MNIST

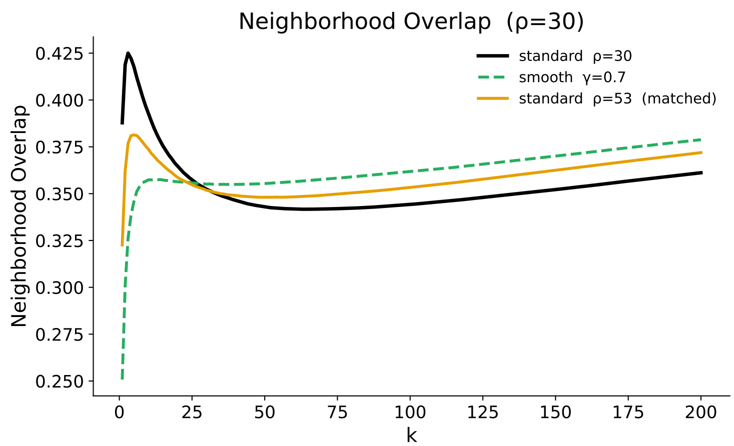


(b) mouse cortex

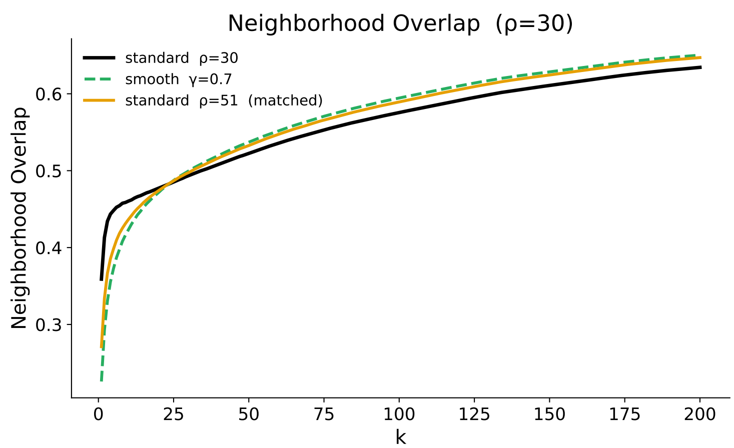


(c) adult

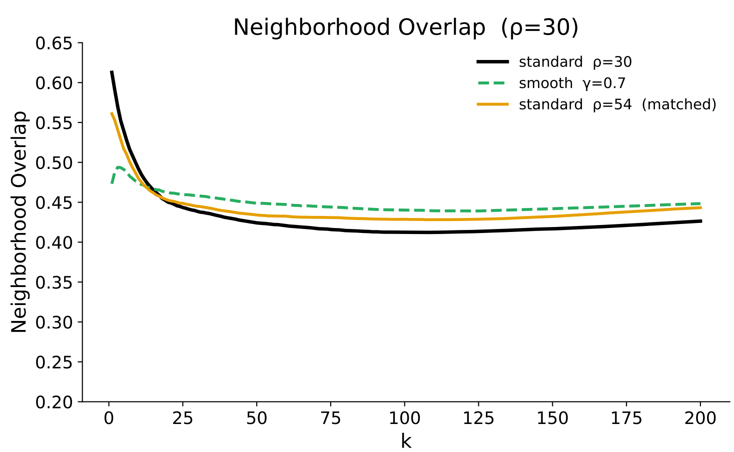
Fig. 3: standard t-SNE versus Smooth t-SNE with $\gamma = 0.5$ across three datasets.



(a) MNIST

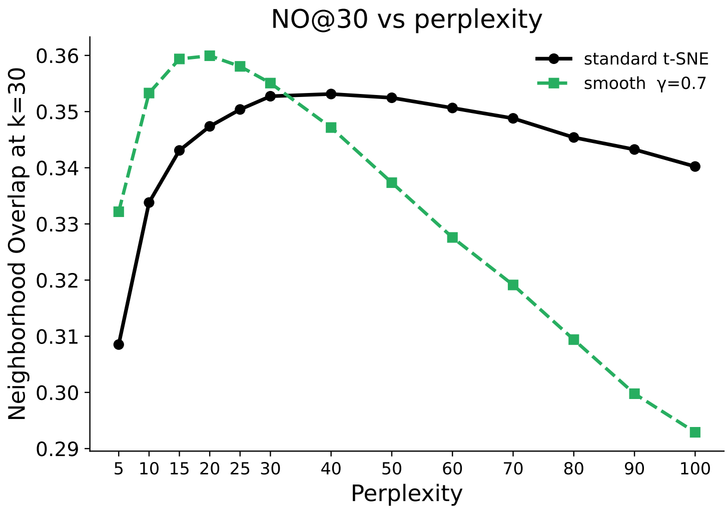


(b) Mouse cortex

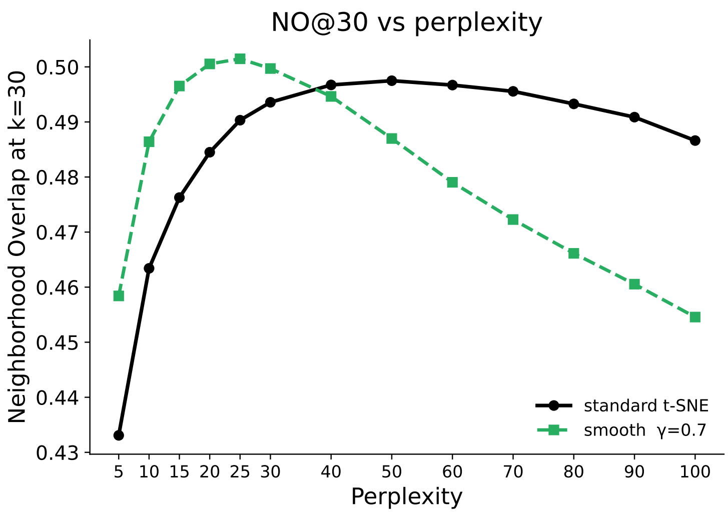


(c) Adult

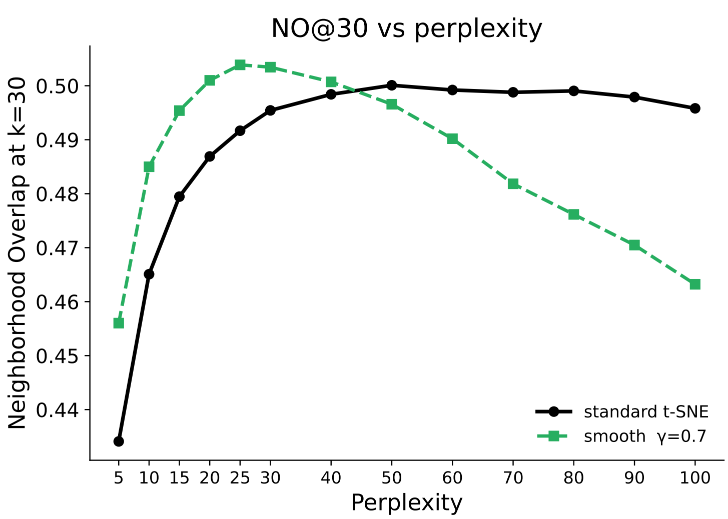
Fig. 4: $NO@k$ comparison between standard t-SNE, smoothed t-SNE ($\gamma = 0.7$), and matched-perplexity standard t-SNE across three datasets.



(a) MNIST

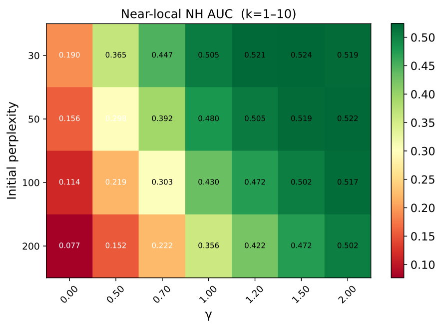


(b) Mouse cortex

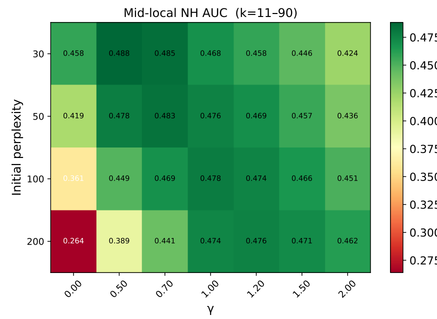


(c) Adult

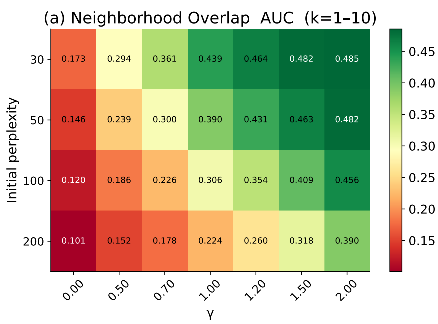
Fig. 5: $NO@k$ comparison between standard t-SNE, smoothed t-SNE ($\gamma = 0.7$),



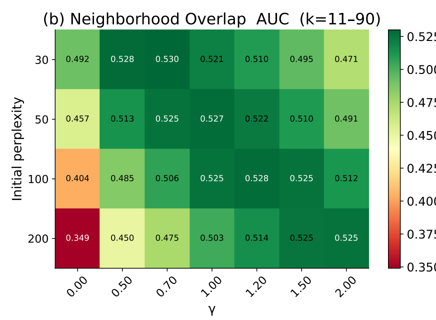
(a) Near-local AUC, MNIST



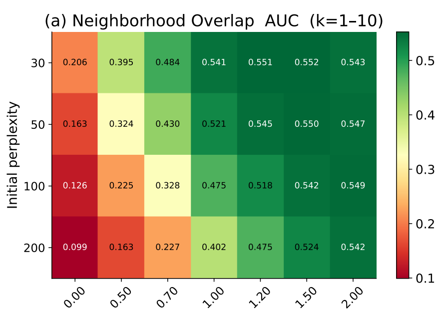
(b) Mid-local AUC, MNIST



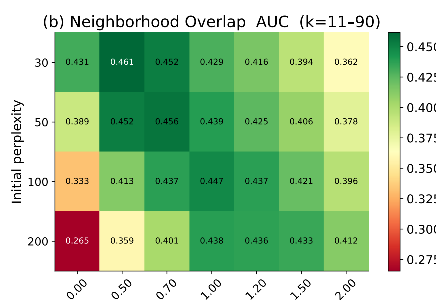
(c) Near-local AUC, Mouse cortex



(d) Mid-local AUC, Mouse cortex

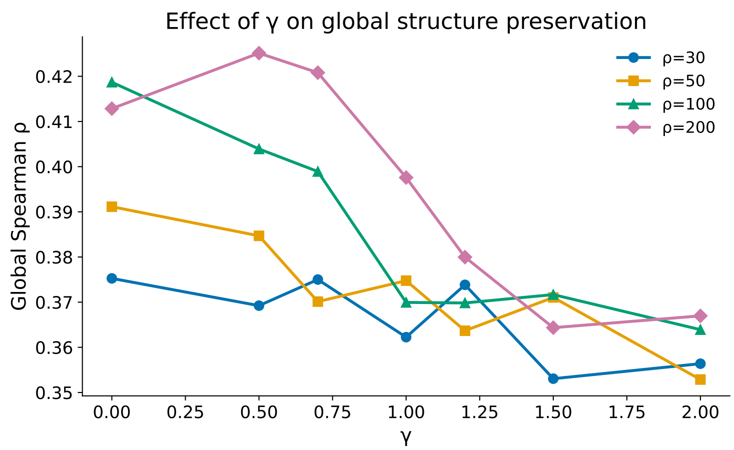


(e) Near-local AUC, Adult

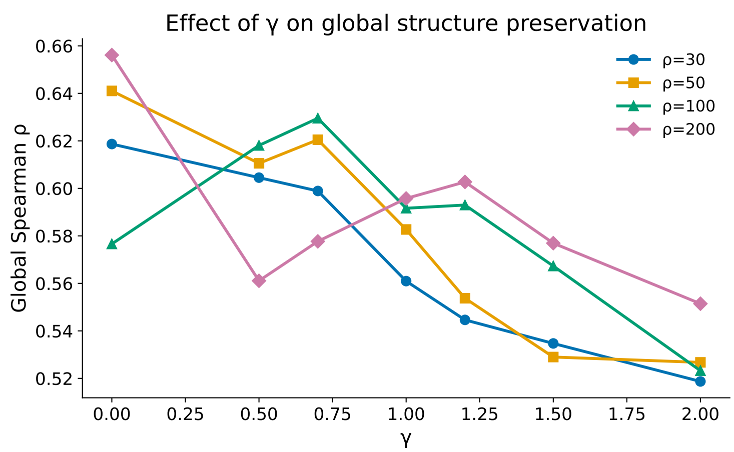


(f) Mid-local AUC, Adult

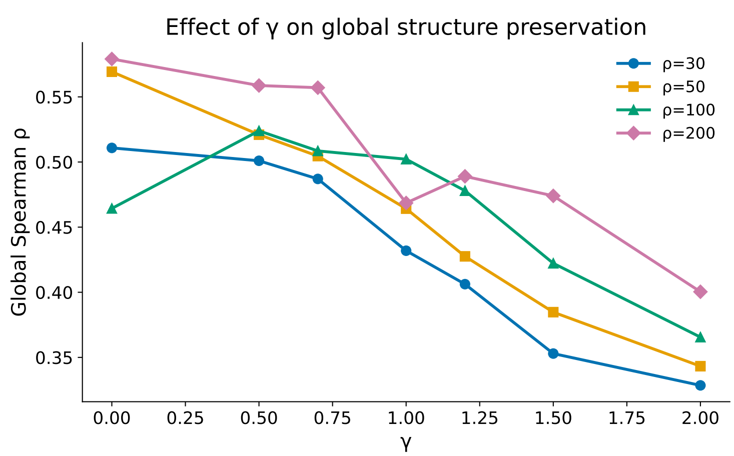
Fig. 6: Sensitivity of neighborhood preservation to γ and ρ across three datasets.



(a) MNIST



(b) Mouse cortex



(c) Adult

Fig. 7: Effect of γ on global structure preservation across three datasets.